

$$\int_V \frac{\partial}{\partial x_i} \left(\frac{\partial \phi}{\partial x_i} \right) dV \rightarrow \int_V \frac{\partial}{\partial x_i} \left(\frac{\partial \phi}{\partial x_i} \right) dV =$$

$$\int_V \frac{\partial}{\partial x_i} (A_{ijkl} \dots) dV = \int_S v_i A_{ijkl} \dots dS,$$

$$\int_V \frac{\partial}{\partial x_i} (\phi v_i) dV = \int_S \underbrace{n_i \phi v_i}_{A_i} = \int_S \underline{v \cdot n} \phi$$

a. $\int_S x_i n_j dS = V \delta_{ij}$

$$B_{ij} = \int_S x_i n_j dS = \int_V \frac{\partial}{\partial x_j} (x_i) dV = \delta_{ij} \int_V dV = V \delta_{ij}$$

$$\underline{\underline{B}} = V \underline{\underline{I}}$$

b. $\int_S \mathbf{n} \times (\mathbf{a} \times \mathbf{x}) dS = 2V \mathbf{a}$



$$b_i = \int_S \epsilon_{ijk} n_j (\mathbf{a} \times \mathbf{x})_k = \int_S \epsilon_{ijk} n_j \epsilon_{kmp} a_m x_p dS =$$

$$= \int_S \epsilon_{ijk} \epsilon_{kmp} n_j a_m x_p dS = \int_S n_p a_i x_p - n_m a_m x_i =$$

$$(\delta_{im} \delta_{jp} - \delta_{ip} \delta_{jm}) n_j a_m x_p = \delta_{im} \delta_{jp} n_j a_m x_p - \delta_{ip} \delta_{jm} n_j a_m x_p$$

$$= a_i \int_S n_p x_p - a_m \int_S n_m x_i = a_i \int_V \frac{\partial x_p}{\partial x_p} - a_m \int_V \frac{\partial x_i}{\partial x_m} =$$

teorema de Gauss

$$= (a_i 3 - a_i) \cdot \int_V dV = 2 a_i V$$

$$b_i = 2 a_i V \Rightarrow \underline{\underline{b}} = 2 V \underline{\underline{a}}$$

c. $\int_S \lambda \mathbf{w} \cdot \mathbf{n} dS = \int_V \mathbf{w} \cdot \nabla \lambda dV$

teorema de Gauss

$$\phi = \int_S (\lambda \mathbf{w}) \cdot \mathbf{n} dS = \int_S (\lambda w_i) n_i dS = \int_V \frac{\partial}{\partial x_i} (\lambda w_i) dV = \int_V \left(\frac{\partial \lambda}{\partial x_i} w_i + \lambda \frac{\partial w_i}{\partial x_i} \right) dV$$

$$\lambda \cdot \frac{\partial w_i}{\partial x_i} = \lambda \frac{\partial}{\partial x_i} \epsilon_{ijk} \frac{\partial v_k}{\partial x_j} = \lambda \epsilon_{ijk} \frac{\partial v_k}{\partial x_i \partial x_j} = \lambda \frac{1}{2} \left(\epsilon_{ijk} \frac{\partial v_k}{\partial x_i \partial x_j} + \epsilon_{ikj} \frac{\partial v_k}{\partial x_i \partial x_j} \right)$$

Prop de la derivada cruzada

$$= \lambda \frac{1}{2} \left(e_{ijk} \frac{\partial v_k}{\partial x_i \partial x_j} - e_{jik} \frac{\partial v_k}{\partial x_i \partial x_j} \right) = \lambda \frac{1}{2} \left(\underbrace{e_{ijk} \frac{\partial v_k}{\partial x_i \partial x_j}} - \underbrace{e_{jik} \frac{\partial v_k}{\partial x_j \partial x_i}} \right) = 0$$

$$\phi = \int_V \underline{\nabla} \lambda \cdot \underline{w} \, dV$$